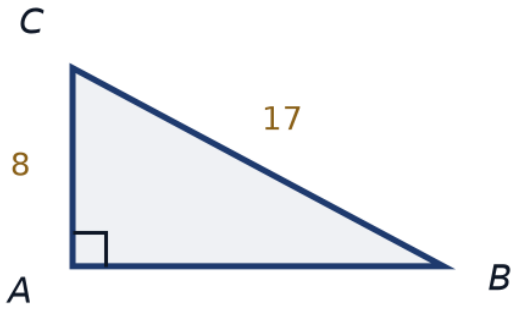
Trigonometry

ClipSAT Printable Practice Suite (clipsat.org) • Topic 6.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

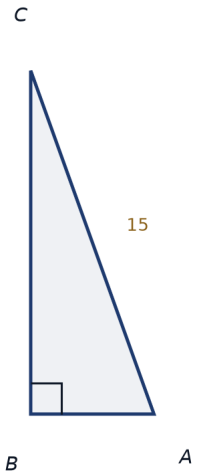
1 [Foundational] • Target: Right triangle trigonometric ratios

In a right triangle, angle θ has an opposite side of 8 and hypotenuse 17. Find $\sin\theta$, $\cos\theta$, and $\tan\theta$.



2 [Foundational] • Target: Right triangle trigonometric ratios

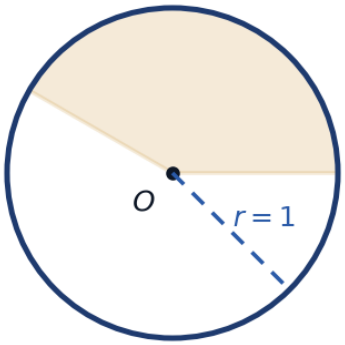
A ladder 15 feet long leans against a wall, making a 70° angle with the ground. Find the height reached on the wall, to the nearest tenth.



© ClipSAT — clipsat.org | All Rights Reserved

3 [Foundational] • Target: The unit circle and radian measure

Convert 150° to radians, in terms of π.

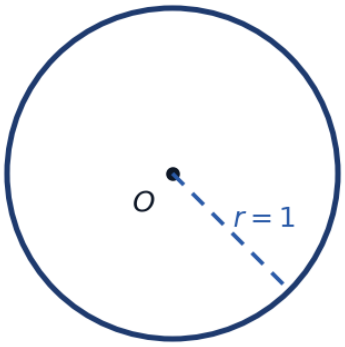


4 [Foundational] • Target: The unit circle and radian measure

Convert $\frac{3\pi}{4}$ radians to degrees.

5 [Foundational] • Target: The unit circle and radian measure

Find the exact value of $\sin\left(\frac{\pi}{6}\right)$ and $\cos\left(\frac{\pi}{4}\right)$, using the unit circle.



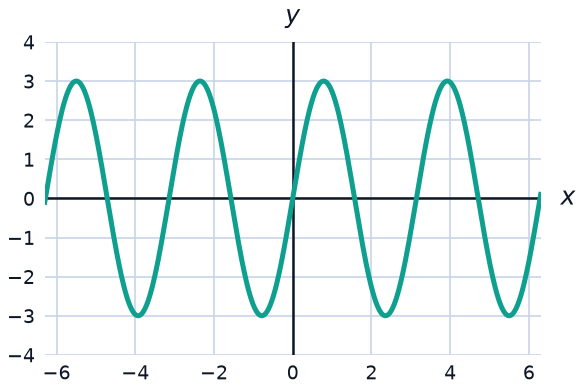
6 [Foundational] • Target: Basic trigonometric identities

Use the Pythagorean identity to find $\cos\theta$, given that $\sin\theta = \frac{3}{5}$ and θ is in the first quadrant.

© ClipSAT — clipsat.org | All Rights Reserved

7 [Foundational] • Target: Graphs of sine and cosine functions

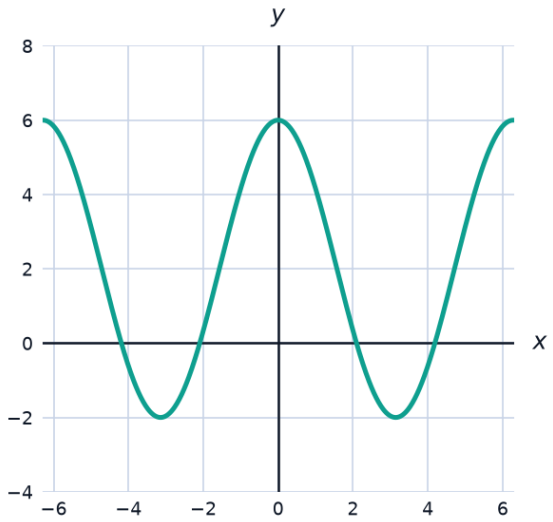
State the amplitude and period of $y = 3\sin(2x)$.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Graphs of sine and cosine functions

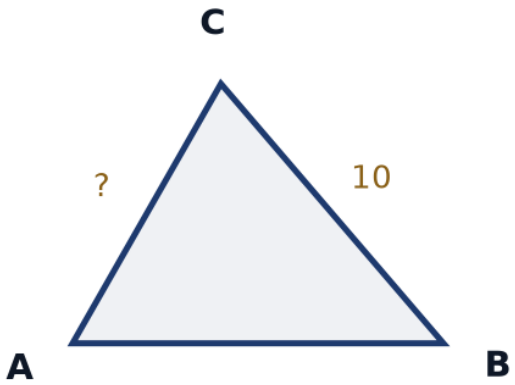
State the amplitude, period, and vertical shift of $y = 4\cos(x) + 2$.



© ClipSAT — clipsat.org | All Rights Reserved

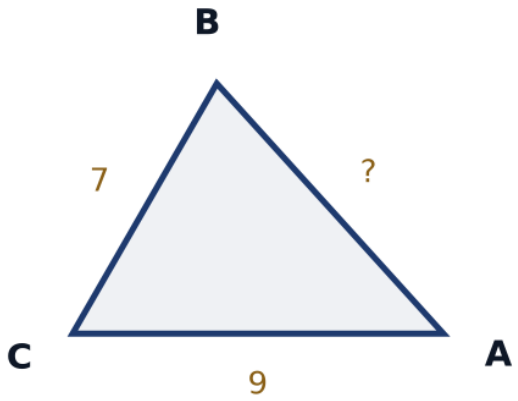
9 [Intermediate] • Target: Law of Sines

In triangle ABC, $A = 40^\circ$, $B = 65^\circ$, and $a = 10$. Find side b, to the nearest tenth, using the Law of Sines.



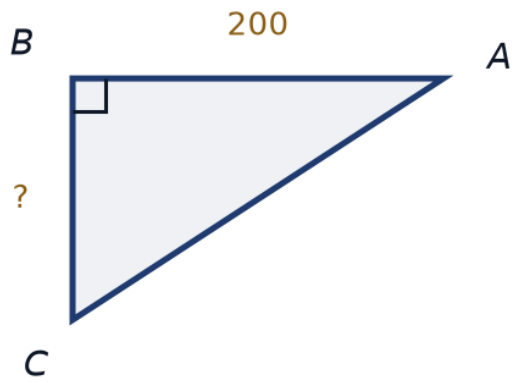
10 [Intermediate] • Target: Law of Cosines

In triangle ABC, $a = 7$, $b = 9$, and $C = 60^\circ$. Find side c, using the Law of Cosines.



11 [Intermediate] • Target: Applications of trigonometry to real-world problems

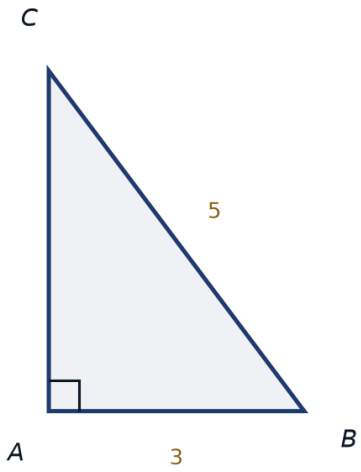
This question has two parts. A surveyor stands 200 feet from the base of a tower and measures the angle of elevation to the top as 38°. (a) Find the height of the tower, to the nearest foot. (b) If the surveyor moves 50 feet closer to the tower, find the new angle of elevation, to the nearest degree, using the height found in part (a).



© ClipSAT — clipsat.org | All Rights Reserved

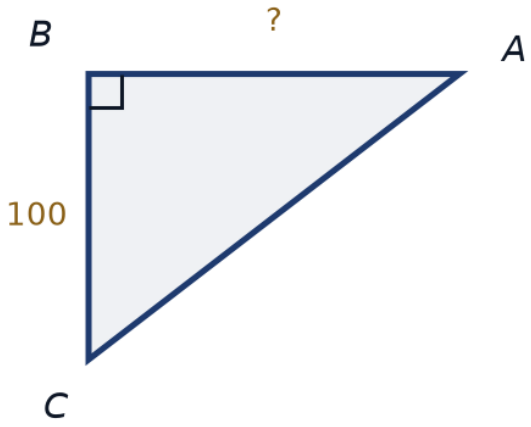
12 **[Intermediate]** • **Target: Reciprocal trigonometric functions**

If $\cos\theta = \frac{3}{5}$, find $\sec\theta$ and $\tan\theta$ (assuming θ is in the first quadrant).



13 **[Intermediate]** • **Target: Angle of elevation and depression**

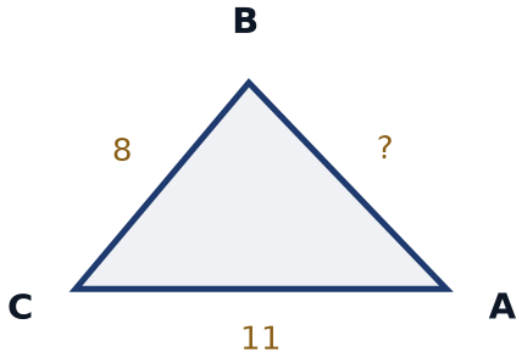
From the top of a 100-foot cliff, the angle of depression to a boat is 25°. Find the horizontal distance from the base of the cliff to the boat, to the nearest foot.



© ClipSAT — clipsat.org | All Rights Reserved

14 [Intermediate] • Target: A multi-part triangle problem combining Law of Sines and Law of Cosines

This question has two parts. Triangle ABC has $a = 8$, $b = 11$, and $C = 50^\circ$. (a) Use the Law of Cosines to find side c, to the nearest tenth. (b) Use the Law of Sines to find angle A, to the nearest degree.



Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Cofunction and negative angle identities • Cross-Lesson: 7.01 Statistics & Probability

Use the cofunction identity to find $\cos(30^\circ)$ given that $\sin(60^\circ) = \frac{\sqrt{3}}{2}$.

16 [Challenging] • Target: Cofunction and negative angle identities • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers

Use the fact that sine is an odd function to find $\sin(-30^\circ)$, given $\sin(30^\circ) = \frac{1}{2}$.

17 [Challenging] • Target: Graphs of sine and cosine functions • Cross-Lesson: 1.01 Numbers, Percents & Ratios

In right triangle ABC with $C = 90^\circ$, $a = 5$, and $c = 13$. What is the value of $\sin A + \cos A$?

18 [Challenging] • Target: Law of Sines • Cross-Lesson: 2.01 Elementary & Intermediate Algebra

Two sides of a non-degenerate triangle measure 7 inches and 11 inches. What is the range of possible perimeters for this triangle?

19 [Challenging] • Target: Law of Cosines • Cross-Lesson: 3.01 Functions & Graphs

What is the period and amplitude of the trigonometric function $f(x) = -6\cos(4x - \pi) + 3$?

20 [Challenging] • Target: Trigonometric equations • Cross-Lesson: 4.01 Coordinate Geometry

A circle has a radius of 12 cm. What is the area of a sector formed by a central angle of 75°?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

© ClipSAT — clipsat.org | All Rights Reserved

21 [Foundational] • Target: Right triangle trigonometric ratios

In a right triangle, the angle θ has an opposite side of length 20, adjacent side 21, and hypotenuse 29. Find $\tan\theta$.

a. $\frac{29}{21}$

b. $\frac{20}{21}$

c. $\frac{29}{20}$

d. $\frac{21}{20}$

22 [Foundational] • Target: The unit circle and radian measure

In triangle ABC, $a = 14$, $b = 7$, and $C = 105^\circ$. Find side c, to the nearest tenth, using the Law of Cosines.

a. 18.7

b. 17.2

c. 20.6

d. 15.7

23 [Foundational] • Target: Basic trigonometric identities

In ΔABC, side AB has length 6, side AC has length 10, and the measure of $\angle BAC$ is 120°. The angle bisector of $\angle BAC$ intersects side BC at point D. What is the length of segment AD?

a. $\frac{15}{4}$

b. $\frac{15\sqrt{3}}{4}$

c. $\frac{15}{2}$

d. $\frac{30}{7}$

24 [Foundational] • Target: Graphs of sine and cosine functions

In ΔABC, the length of side AC is 5, the length of side BC is 8, and the measure of $\angle C$ is 60°. What is the value of $\cos A$?

a. $\frac{1}{7}$

b. $\frac{2}{7}$

c. $\frac{5}{8}$

d. $\frac{11}{14}$

25 [Foundational] • Target: Law of Sines

In a right triangle, the side opposite θ is 8, the side adjacent is 6, and the hypotenuse is 10. Find $\tan\theta$.

a. 0

b. $\frac{3}{4}$

c. $\frac{4}{3}$

d. $\frac{2}{25}$

© ClipSAT — clipsat.org | All Rights Reserved

26 [Foundational] • Target: Law of Cosines

In a triangle, sides $a = 15$ and $b = 14$ meet at an included angle of 123°. Using the Law of Cosines, which expression gives side c?

a. $\sqrt{15^2 + 14^2 + 2(15)(14)\cos 123^\circ}$

b. $\sqrt{15^2 - 14^2 - 2(15)(14)\cos 123^\circ}$

c. $\sqrt{15^2 + 14^2 - 2(15)(14)\cos 123^\circ}$

d. $15^2 + 14^2 - 2(15)(14)\cos 123^\circ$

27 [Foundational] • Target: Trigonometric equations

In a right triangle, the side opposite angle θ is 18, the adjacent side is 24, and the hypotenuse is 30. What is the sine of θ?

a. $\frac{3}{4}$

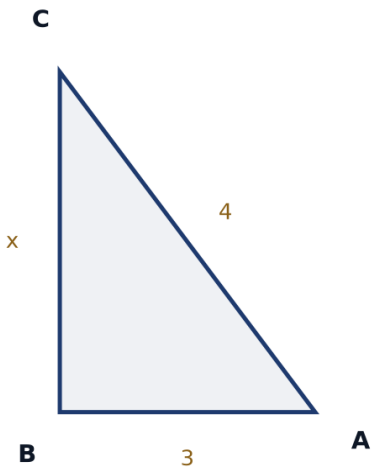
b. $\frac{5}{3}$

c. $\frac{4}{5}$

d. $\frac{3}{5}$

28 [Foundational] • Target: Applications of trigonometry to real-world problems

In the right-angled triangle ABC shown below, $\angle ABC = 90^\circ$, $AB = 3$, and $BC = 4$. What is the length of the hypotenuse AC?



a. 5

b. 7

c. 12

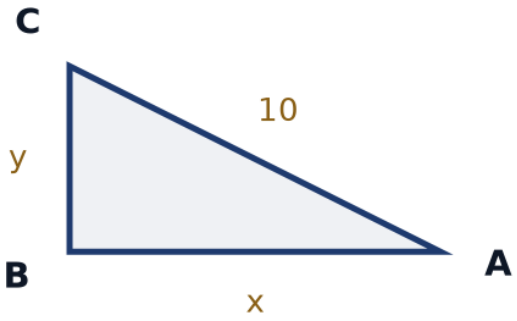
d. 25

© ClipSAT — clipsat.org | All Rights Reserved

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Reciprocal trigonometric functions**

In the right-angled triangle shown, what is the length of side AC if $AB = 10$ and $\angle B = 30^\circ$?



a. 5

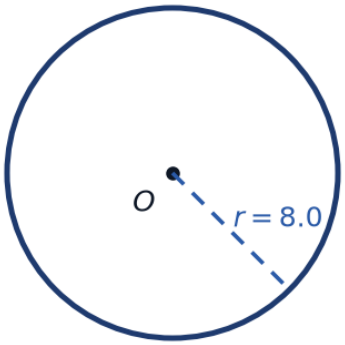
b. 10

c. $5\sqrt{3}$

d. 20

30 **[Intermediate]** • **Target: Angle of elevation and depression**

A circle with radius 8 has a sector with area 16π. What is the measure of the central angle of this sector in radians?



a. $\frac{\pi}{4}$

b. $\frac{\pi}{2}$

c. $\frac{2\pi}{3}$

d. π

© ClipSAT — clipsat.org | All Rights Reserved

31 [Intermediate] • Target: Solving trigonometric equations over a restricted domain

What is the reference angle of 140°?

a. 64°

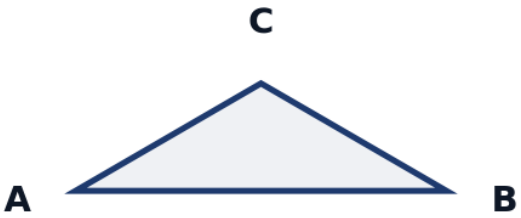
b. 40°

c. 66°

d. 31°

32 [Intermediate] • Target: A multi-part triangle problem combining Law of Sines and Law of Cosines

In a right triangle, the hypotenuse is 10 and one angle is 30°. What is the length of the side opposite the 30° angle?



a. 6

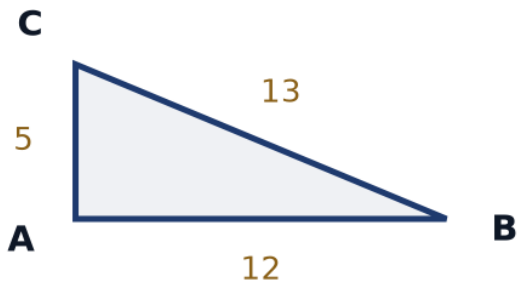
b. 5

c. 10

d. 4

33 [Intermediate] • Target: Cofunction and negative angle identities

A right triangle has legs 5 and 12 with hypotenuse 13. Find $\tan\theta$, where θ is the angle opposite the side of length 5.



a. $\frac{12}{13}$

b. $\frac{13}{5}$

c. $\frac{12}{5}$

d. $\frac{5}{12}$

© ClipSAT — clipsat.org | All Rights Reserved

34 [Intermediate] • Target: Right triangle trigonometric ratios

If θ is an angle in Quadrant III, what is the sign of $\tan\theta$?

a. positive

b. undefined

c. negative

d. zero

35 [Intermediate] • Target: The unit circle and radian measure

If θ is an angle in Quadrant II, what is the sign of $\tan\theta$?

a. undefined

b. negative

c. zero

d. positive

36 [Intermediate] • Target: Basic trigonometric identities

If θ is an angle in Quadrant IV, what is the sign of $\sin\theta$?

a. negative

b. zero

c. undefined

d. positive

37 [Intermediate] • Target: Graphs of sine and cosine functions

If θ is an angle in Quadrant I, what is the sign of $\tan\theta$?

a. positive

b. undefined

c. negative

d. zero

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Law of Sines • Cross-Lesson: 2.01 Elementary & Intermediate Algebra

In a right triangle, angle A and angle B are acute angles such that $\sin(A) = \frac{7}{11}$. What is the value of $\cos(B)$?

a. $\frac{7}{11}$

b. $\frac{4}{11}$

c. $\frac{8}{11}$

d. $\frac{11}{7}$

© ClipSAT — clipsat.org | All Rights Reserved

39 [Challenging] • Target: Law of Cosines • Cross-Lesson: 3.01 Functions & Graphs

What is the domain of the real-valued function $f(x) = \sqrt{2x - 14}$?

a. $x \geq -7$

b. $x \leq 7$

c. $x > 7$

d. $x \geq 7$

40 [Challenging] • Target: Trigonometric equations • Cross-Lesson: 4.01 Coordinate Geometry

What is the radian measure of an angle of 330°?

a. $\frac{11\pi}{6}$

b. $\frac{\pi}{33}$

c. $\frac{330\pi}{360}$

d. $\frac{330}{180\pi}$

41 [Challenging] • Target: Applications of trigonometry to real-world problems • Cross-Lesson: 5.01 Plane Geometry

In a right triangle ABC where angle $C = 90^\circ$, if $\cos(A) = \frac{12}{13}$, what is the value of $\tan(A)$?

a. $\frac{5}{12}$

b. $\frac{12}{5}$

c. $\frac{5}{13}$

d. $\frac{13}{12}$

42 [Challenging] • Target: Reciprocal trigonometric functions • Cross-Lesson: 7.01 Statistics & Probability

A point $P(x, y)$ on the unit circle corresponds to an angle of θ. If $\cos(\theta) = -\frac{3}{5}$ and θ is in Quadrant II, what is $\sin(\theta)$?

a. $-\frac{4}{5}$

b. $\frac{4}{5}$

c. $-\frac{3}{4}$

d. $\frac{3}{4}$

43 [Challenging] • Target: Angle of elevation and depression • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers

An inscribed angle in a circle intercepts an arc with a measure of 110°. What is the measure of the inscribed angle?

a. 45°

b. 55°

c. 70°

d. 110°

© ClipSAT — clipsat.org | All Rights Reserved

44 [Challenging] • Target: Solving trigonometric equations over a restricted domain • Cross-Lesson: 1.01 Numbers, Percents & Ratios

A right triangle has legs of length $5\sqrt{3}$ and 5. What is the measure of the smallest angle in degrees?

a. 30°

b. 45°

c. 60°

d. 15°

45 [Challenging] • Target: A multi-part triangle problem combining Law of Sines and Law of Cosines • Cross-Lesson: 2.01 Elementary & Intermediate Algebra

What is the measure of each interior angle of an equilateral triangle?

a. 45°

b. 60°

c. 75°

d. 90°

© ClipSAT — clipsat.org | All Rights Reserved